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Olive oil sensory defects classification with data fusion of instrumental techniques and multivariate analysis (PLS-DA)

Eva Borràs [a], Joan Ferré [b], Ricard Boqué [b], Montserrat Mestres [a], Laura Aceña [a], Angels Calvo [c], Olga Busto [a,*]

[a] iSens Group, Department of Analytical Chemistry and Organic Chemistry, Universitat Rovira i Virgili, Campus Sescelades, 43007 Tarragona, Spain
[b] Chemometrics, Qualimetrics and Nanosensors Group, Department of Analytical Chemistry and Organic Chemistry, Universitat Rovira i Virgili, Campus Sescelades, 43007 Tarragona, Spain
[c] Official Taste Panel of Virgin Olive Oil in Catalonia, Reus, Spain

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ABSTRACT

Three instrumental techniques, headspace-mass spectrometry (HS-MS), mid-infrared spectroscopy (MIR) and UV–visible spectrophotometry (UV–vis), have been combined to classify virgin olive oil samples based on the presence or absence of sensory defects. The reference sensory values were provided by an official taste panel. Different data fusion strategies were studied to improve the discrimination capability compared to using each instrumental technique individually. A general model was applied to discriminate high-quality *non-defective* olive oils (extra-virgin) and the lowest-quality olive oils considered *non-edible* (lampante). A specific identification of key off-flavours, such as musty, winey, fusty and rancid, was also studied. The data fusion of the three techniques improved the classification results in most of the cases. Low-level data fusion was the best strategy to discriminate musty, winey and fusty defects, using HS-MS, MIR and UV–vis, and the rancid defect using only HS-MS and MIR. The mid-level data fusion approach using partial least squares-discriminant analysis (PLS-DA) scores was found to be the best strategy for *defective vs non-defective* and *edible vs non-edible* oil discrimination. However, the data fusion did not sufficiently improve the results obtained by a single technique (HS-MS) to classify *non-defective* classes. These results indicate that instrumental data fusion can be useful for the identification of sensory defects in virgin olive oils.

1. Introduction

Virgin olive oil (VOO) is a valuable and highly appreciated vegetable oil. It is extracted from fresh and healthy olive fruits (*Olea europeae* L.) by mechanical and other physical methods without additional refining (Cecchi & Alfei, 2013). Due to its unique nutritional and organoleptic properties there is an increasing interest of consumers for olive oil quality. Olive oil sensory and chemical quality characteristics depend on olive variety, environmental factors, agronomic techniques and cultivation, production and storage conditions (Bendini, Valli, Barbieri, & Gallina Toschi, 2012; García-González & Aparicio, 2010b; Karabagias et al., 2013). Furthermore, oils of higher quality can be intentionally blended with cheaper vegetable oils or olive oil grades, which constitute, in addition to economic fraud, a health threat to the consumer. To guarantee VOO quality different international institutions (ECC/2568/1991; ECC/796/2002, ECC/1089/2003 and ECC/640/2008; International Olive Council, 2013 and the Codex Alimentarius (FAO-OMS, 1981)) have accorded maximum values of specific parameters that cannot be exceeded by each olive oil category.

Three physico-chemical parameters (free acidity, peroxide value and specific ultraviolet (UV) absorptions) and a sensorial evaluation (based on taste and aroma) determine the three quality categories of VOOs: (1) extra-virgin olive oil (EVOO), which has the maximum quality and no sensory defects; (2) virgin olive oil (VOO), which may have some negative sensory attributes if they are of low intensity ($\leqslant$3.5); and (3) and lampante olive oil (LOO), with intense defects (>3.5). LOO cannot be directly consumed and must undergo a prior refining process, while EVOOs and VOOs can be bottled and directly consumed. From an economic point of view, it is important to know how an olive oil is qualified through the presence of off-flavours (Monteleone & Langstaff, 2014), discriminating the lower-quality category (LOO), called *non-edible* from the highest quality olive oils (EVOO), called *non-defective* (Escuderos, García, Jiménez, & Horrillo, 2013). The only homologated method to assess the sensory attributes of olive oils is the evaluation by an official taste panel, following well-standardized

* Corresponding author.
E-mail address: olga.busto@urv.cat (O. Busto).

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protocols with highly and permanently trained panelists, which evaluate the sight, aroma, taste, texture and aftertaste of the oil (Monteleone, 2014). The sensory descriptors of olive oil are classified as 'positive attributes', such as fruity, bitter and pungent, and 'negative' attributes. The latter describes the defects of olive oil, and include fusty (along with muddy sediment (Purcaro, Cordero, Liberto, Bicchi, & Conte, 2014)), musty-humidity, winey-vinegary, rancid and metallic. An intense process of auto oxidation caused by unsuitable storage conditions is the main cause of rancidity in olive oils, whereas the presence of exogenous enzymes, mostly due to microbial activity is mainly responsible for musty (by fungi and yeast), fusty and winey-vinegary defects (Morales, Luna, & Aparicio, 2005).

Sensory descriptors mainly depend on the content of volatile and non-volatile minor components. Non-volatile compounds, such as phenolic compounds, stimulate the taste receptors, evoking the perception of bitterness, pungency, astringency and metallic attributes. Volatile compounds, mainly arising from the oxidation of the fatty acids, stimulate the olfactory receptors, responsible for the whole VOO aroma (Angerosa, 2002). This combined effect of taste, odour and chemical interactions, perceived as 'flavour', is well-established by the sensory human panel. However, this methodology has some inherent problems, such as experts' subjectivity, variability of responses over time, the lack of reference standards and low number of analyses per day (Sinelli et al., 2010).

The development of an objective, rapid, automated, low-cost and precise methodology to assess sensory properties using instrumental techniques might be an alternative solution to taste panels. Several approaches have been proposed as alternatives to human sensory panels to evaluate the quality of olive oil. The first attempts focused on the characterization of the chemical species responsible for the flavour, by correlating negative and positive attributes to specific responses of chemical compounds, such as volatile and phenolic compounds. Volatile compounds, which are responsible for the aroma, were initially studied with traditional electronic-noses based on gas sensor arrays. Metal-oxide sensors (MOS) were applied to find relationships between organoleptic descriptors and sensor responses (Aparicio, Rocha, Delgadillo, & Morales, 2000; Escuderos et al., 2013). However, due to the lack of sensitivity and selectivity of the sensor devices, electronic-noses based on mass spectrometry gained popularity. Different methods have been used to pre-concentrate the volatile fraction: static headspace (HS), dynamic headspace (DHS) (Aparicio, Luna, & Morales, 2000; Morales et al., 2005) and solid-phase micro-extraction (HS-SPME) (Purcaro et al., 2014; Vichi, Romero, Tous, Tamames, & Buxaderas, 2008) providing a characteristic fingerprint of the volatile compounds. The most common strategy is the combination of gas chromatography (GC) and a pre-concentration method. Another option to analyze volatile compounds is by directly coupling a headspace system to a mass spectrometer (HS-MS) without performing a chromatographic separation. Although the characterization of the volatile compounds is not achieved, the global volatile 'spectral fingerprint' is considered to be correlated with the main sensory defects (López-Feria, Cárdenas, García-Mesa, & Valcárcel, 2008).

The non-volatile compounds, like phenolic compounds, have also been studied using electronic-tongues based on liquid chemical sensor arrays (Apetrei et al., 2010). Nevertheless, problems, such as low conductivity, viscosity and limited solubility of the oil samples, make the liquid sensors a complex system to work with. Vibrational spectroscopy, such as near- and mid-infrared (NIR and MIR), is an alternative to the liquid sensor systems due to its simplicity, rapidness and affordability. NIR and MIR spectra offer significant 'fingerprint' information about individual components in complex samples that can be connected to the sensory attributes described by the human panel (Sinelli et al., 2010). Related to the colour measurements, another alternative fingerprint technique is ultraviolet–Visible spectrophotometry (UV–vis). Although the colour is not a descriptor considered by the official taste panel, it may influence the quality of the olive oil. Few works have been conducted to study the colour sensory descriptors (Kružlicová, Mocák, Katsoyannos, & Lankmayr, 2008), and have mainly focused on region authentication (Pizarro, Rodríguez-Tecedor, Pérez-del-Notario, Esteban-Díez, & González-Sáiz, 2013) and time evolution (Tarakowski, Malanowski, Kościesza, & Siegoczyński, 2014).

Panelists perceive sensory attributes and defects of olive oils as a mixture of gustatory, olfactory and tactile perceptions. These particular sensory properties are due to the presence of several compounds that may interact through synergisms and antagonisms. So, for example, one single compound may contribute to different sensory properties and at the same time one single property may depend on many compounds for explanation. For this reason, to model the sensory parameters, a combination of different instrumental techniques is required, which provide complementary information from different sources to make sensorial description easier. To process the data suitable multivariate pattern recognition techniques are required, especially from non-selective spectral data. The combination of the information from different instruments is called data fusion and essentially it is carried out at three levels. The low-level data from different instruments are simply concatenated ('fused') into a single matrix prior to multivariate modeling. Mid-level data fusion first extracts features from the individual matrices and then fuses the new variables into a 'scores-matrix' which is used to build the final model. The new features are usually extracted by Principal Component Analysis (PCA) or partial least squares-discriminant analysis (PLS-DA). Finally, high-level fusion multivariate models are built on each individual technique and the individual predictions are combined to produce the final result (Borràs et al., 2015). Some data fusion approaches have been proposed to assess olive oil quality by fusing responses from electronic sensor devices, such as gas sensors (MOX or MOS) and liquid sensors (Apetrei et al., 2010), from spectroscopic measurements, such as MIR and NIR (Dupuy, Galtier, Ollivier, Vanloot, & Artaud, 2010), mass spectrometry (MS) with optical instruments, such as UV–vis spectrophotometry, (Casale, Armanino, Casolino, & Forina, 2007) or CIELab colour space, chemical parameters and combinations of them (Pizarro et al., 2013). Most of the studies cited were applied to the authentication of olive oil samples according to their origin and only a few studies applied data fusion to correlate human sensory responses (Apetrei et al., 2010).

This study evaluates the feasibility of combining an electronic nose based on headspace mass spectrometry (HS-MS), an electronic tongue based on MIR spectroscopy and an electronic eye based on UV–vis spectrophotometry to discriminate the main olive oil categories (extra-virgin, virgin and lampante) and to detect the main off-flavours (musty, winey-vinegary, fusty and rancid) of olive oils. The instrumental data were correlated to the sensory results using PLS-DA to finally find the best data fusion strategy for improving the discriminating capability of the techniques. For this purpose olive oil samples from the Catalonia region with the presence of a variety of defects were evaluated by an official taste panel and also analyzed by the three instrumental techniques.

2. Materials and methods

2.1. Olive oil samples

A total of 146 samples were supplied by the 'Official Taste Panel of Virgin Olive Oil in Catalonia' in Reus (Government of Catalonia, Spain) during the seasons 2012 and 2013. Samples were stored in dark bottles at −20 °C under nitrogen atmosphere until instrumental analysis. All of the samples were analyzed within three months after the sensory analysis. Based on the sensory results provided by the panel, 84 were graded as EVOO or *non-defective*, 48 as VOO and 14 as LOO or *non-edible*. 30–40% of the olive oils presented the musty, winery or fusty defects, and only around 10% had the rancid defect.

2.2. Sensory analysis

Sensory analysis was carried out by the official panel following the official method of the Olive Oil Council (COI/T20/Doc15) and within the framework of ECC regulation 640/2008. The panel is also accredited according to the ISO17025 norm. 15 ml of each sample was tasted in a normalized coloured cup to mask the colour differences. The temperature of the oils was kept at 28 ± 2 °C. Samples were labeled with a digital code and served following a balanced rotation plan.

The positive scored attributes were fruitiness, bitterness, pungency, grassy green, astringent, sweet and apple. The scored sensory attributes indicating defectiveness or unpleasantness were musty, winey-vinegary, fusty, rancid and metallic. The samples tested did not show the metallic attribute, and for this reason this defect was not studied. Descriptors were evaluated on a continuous, unlabeled, intensity scale (10 cm), and then transformed into numeric variables between 0 and 10. Each sample was tested by 8–10 panelists and the median value was provided for the predominant attributes.

2.3. Instrumental analysis

The volatile composition was determined by a headspace-mass spectrometry based electronic nose (HS-MS). The volatiles were extracted and concentrated using a solid phase micro-extraction (SPME) fibre in the samples' headspace and detected with a HP5973N Mass Selective Detector. All of the extractions were made by an Autosampler CTC-PAL (Agilent) with divinylbenzene/car boxen/poly-dimethylsiloxane (DVB/CAR/PDMS) SPME fibres of 50/30 μm film thickness and 2 cm length. Olive oil samples (5 g) were weighed and placed into a 20 ml vial closed with a PTFE/silicone septum. Before extraction, the stabilization of the headspace in the vial was reached by equilibration for 30 min at 40 °C. Then the fibre was exposed at 40 °C for 1 h with stirring. After extraction, 5 min of thermal desorption of the fibre was sent into the GC system. To transfer the volatiles from the fibre to the MS detector, a HP-5MS column (30 m × 0.25 mm × 0.25 μm) on a 6890 N gas chromatograph was kept at the suitable temperature to transfer the volatiles to the MS and avoid chromatographic separation. Mass spectra were obtained in the electron impact mode (70 eV) in a mass range from 50 to 350 amu. Two replicates were measured per sample.

To analyze the liquid composition, Fourier-transform mid infrared spectra (MIR) were measured on an FT-MIR Nexus (Thermo Nicolet, USA) spectrometer equipped with a deuterated triglycine sulphate (dTGS) detector. The spectra were collected at room temperature over the range 4000–600 cm^{-1}, at 4 cm^{-1} resolution and with 36 scans both for background and samples. A thick film of each oil sample was homogenously placed over the ZnSe crystal ARK multi-bounce (horizontal attenuated total reflectance, HATR) with 12 reflections. The OMNIC software version 6.2 from Thermo Nicolet was used for spectral acquisition, instrument control and preliminary file manipulation. The spectra were compensated to eliminate interfering H_2O and CO_2 bands by running a blank with air. After that only the intervals with informative signals were selected for modeling, namely the ranges 3257.3–2604.3 cm^{-1} and 1951.4–682.8 cm^{-1}. Three replicates were measured per sample.

The colour of the samples was determined by a single beam UV-Visible Heλios Gamma spectrophotometer (Thermo) equipped with diode array technology. The radiation source was a combination of a deuterium-discharge and a tungsten lamp for the ultraviolet and visible wavelength range respectively. The instrument was controlled by a compatible PC equipped with Vision Thermo software. Spectral data were collected at room temperature within the wavelength range 300–1000 nm at 2 nm resolution. It is referred to as 'UV–Vis spectra', despite acquired data also contained a small region of the near-infrared (NIR) (800–1000 nm) region. All samples were analyzed in duplicate in a quartz cuvette with a path length of 10 mm.

2.4. Statistical data analysis

2.4.1. General

Chemometric models were applied to the experimental data collected by the different instrumental techniques. The main purpose was to classify the olive oils with respect to their sensory defects presence using individual and fused data model strategies.

Preliminary PCA models of each instrumental technique data and for each data fusion approach were studied (Kozak & Scaman, 2008). Afterwards, different modeling strategies using PLS-DA were considered. PLS-DA is a discrimination technique useful when data matrices have more objects than variables. A regression model is calculated relating the independent variables X to an integer Y that designates the class of the sample with a binary response (called 'dummy' matrix). The $\boldsymbol{X}$ matrix (instrumental signals) contained the spectra in rows and Y was a vector with zeroes (for negative, i.e. non-defective) and ones (positive, i.e. defective oils). The class of each sample is determined from the predictions of the PLS model (ranging from zero to one) according to a threshold value that delimits the classes (Bevilacqua, Bucci, Magrì, Magrì, & Nescatelli, 2013).

2.4.2. Optimal model selection

To find the optimal classification model for each class studied, different spectral regions (variables) and combinations of them were considered along with different pre-processing options. The classification models were built with a training set using leave-one-out cross-validation (CV) and the lowest classification error was the criterion used to select the optimal number of PLS-DA latent variables. The final models' performance was confirmed by a test set validation (TV). For that purpose, the dataset was randomly split into a training and test set, with 65% and 35% of the samples, respectively. The split between training and test was done by keeping the ratio of samples of each class like in the original set. The random split procedure between training and test sets was repeated ten times and the mean of the classification parameters (sensitivity, specificity and misclassification rate) of the ten models was used as the final result, thus avoiding results depending on a particular split. The mean values and standard deviations were reported as percentages.

The quality of the models was assessed from the classification and prediction abilities. The optimal conditions were decided by means of sensitivity, specificity and inaccuracy. Sensitivity and specificity are defined as the number of samples correctly classified as negatives (non-defectives) or positives (defectives), respectively. To select the best sensitivity and specificity results a multicriteria Pareto's decision method was applied, where each model was represented in a bidimensional scatter plot reporting sensitivity and specificity on each axis, respectively. A compromise between high sensitivity and specificity was used to find the optimal solutions in the Pareto diagram (Oliveri et al., 2014). Minimum inaccuracy values were also considered. Inaccuracy is defined as the probability of samples from both classes being wrongly classified and it is useful to avoid artificially high errors when classes are unbalanced. The final selection criterion was defined considering inaccuracy and sensitivity/specificity Paretos' multicriteria values for the test-validation results.

2.4.3. Individual and data fusion strategies

In order to find the best combinations to classify the olive oil samples three modeling strategies were tested: individual techniques, and low- and mid-level data fusion approaches (summarized in Fig. 1).

Before the data fusion, individual data from HS-MS (MS), MIR and UV–vis analyses were processed, giving three different fingerprints per sample. The strategy described in Fig. 1a was followed in order to find the most representative variables (regions) and preprocesses for each model. Each spectral matrix was split into a training/test set (65/35) before building PLS-DA models.

In the low-level (LL) fusion approach (Fig. 1b), raw data from MS, MIR and UV–vis were concatenated (joined together in a single matrix) before model calculation. Scaling methods were tested considering the different scale of each data type (especially for the MS). Scaling may be done separately for each data block, but also between blocks. After that, the data fused matrix was randomly split into a training/test set (65/35) to build several PLS-DA models.

In the mid-level (ML) fusion approach (Fig. 1b), relevant features were extracted from the different data blocks and were concatenated into a single matrix. This reduced the dimensionality and allowed each block to be treated individually. Before that, individual matrices were randomly split into a training/test set (65/35). Scores from PCA or PLS-DA models obtained independently from each data matrix were then fused and named MLpca and MLpls respectively (Louw et al., 2009). These new variables selected, which kept enough original information, were subsequently combined to build a 'score-matrix', which was preprocessed by different scaling methodologies before building the final PLS-DA model. To extract the possible features, different strategies were chosen. For PCA features the number of principal components (PCs) was chosen in order to give more than 90% of cumulative variance. To extract PLS-DA features, the number of latent variables (LVs) was chosen when the classification error obtained by cross-validation was minimized. The challenge was to find the optimal combination of extracted features and preprocessing methods that provided the best model. However, examining all of the combinations can make the process cumbersome and computationally intense. In both cases, CV was applied to each training set to decide the optimal number of latent variables or scores used for the new 'score-matrices'. Several final PLS-DA models were applied to the fused score-matrices starting from the training-test set split procedure.

The different strategies were performed using two techniques (MS + MIR), called data fusion 2 (2LL, 2MLpca and 2MLpls); and three techniques (MS + MIR + UV–vis), called data fusion 3 (3LL, 3MLpca and 3MLpls). This is because the official taste panel does not evaluate visual characteristics of the olive oil, so the UV–vis technique might not provide enough information. At this stage, no attempts were made to combine the results of the models (high-level data fusion).

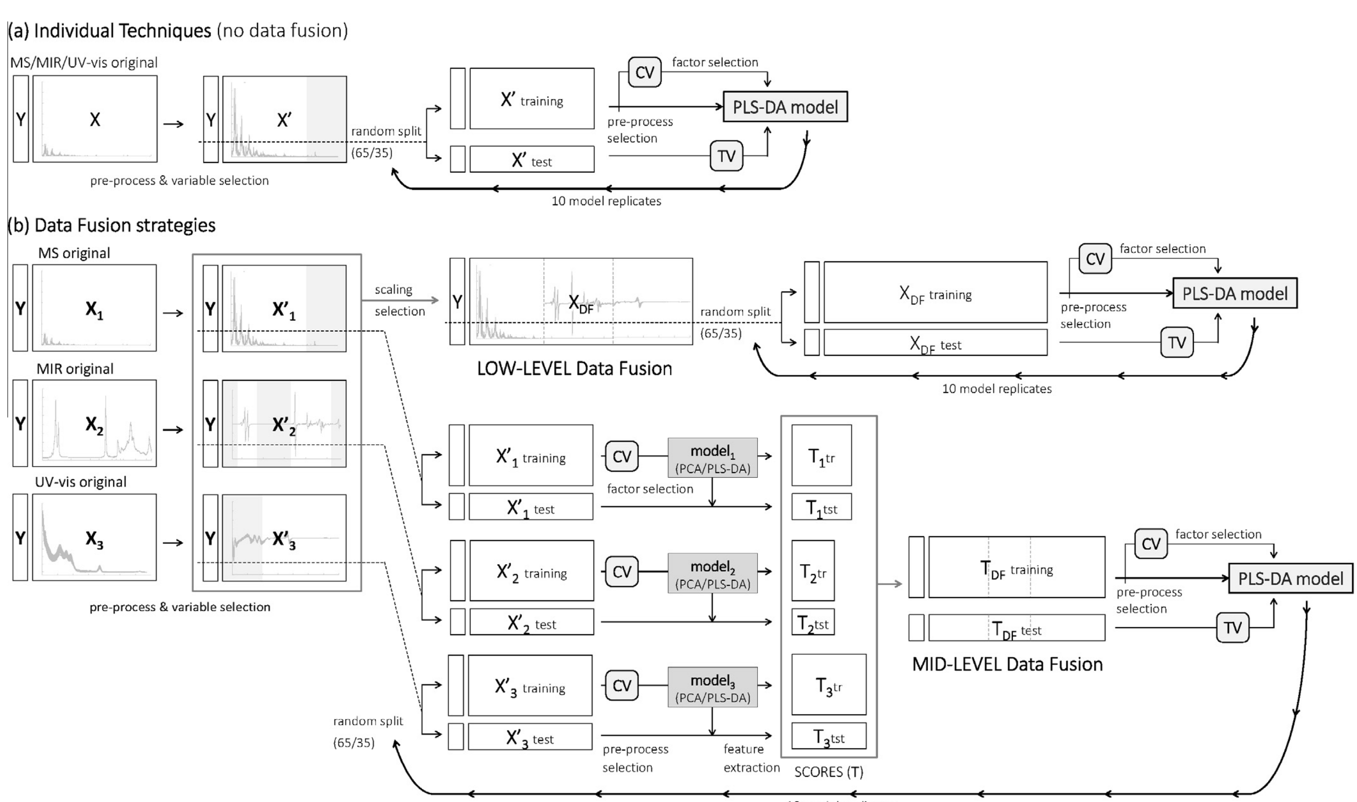

Fig. 1. Individual (a) and data fusion (b) procedures followed to obtain the final optimal PLS models.

All of the data processing was carried out with in-house Matlab v.7.8 routines (Mathworks, MA, USA) and PLS Toolbox software v.6.2 (Eigenvector Research, Manson, WA, USA).

3. Results and discussion

3.1. Individual data results

3.1.1. General

In the first stage, individual headspace/mass spectrometry (MS), mid-infrared (MIR) and UV–visible (UV–vis) spectroscopy data were treated to build the discrimination models. The measurements (Fig. 2) were organized in matrices of dimensions 146×301 (MS), 146×594 (MIR) and 146×701 (UV–vis).

To check the repeatability of the measurements, detect outliers and recognize patterns in the samples' distribution (harvest year, analysis day or other attribute's presence) PCA was performed for each technique. No clear patterns were observed. Considering the measured spectra, six samples for MIR and five samples of both, MS and UV–vis, were removed based on Hotelling's T^2 values and Q residuals well above 95% of the confidence limits.

Then separate PLS-DA models were studied for each instrument. The best pre-process and region (variables), together with the optimal configuration of the PLS-DA models, such as the number of latent variables retained, were selected as those leading to the lowest inaccuracy and highest sensitivity and specificity obtained with the validation set (TV). The optimal conditions for each discrimination model are summarized in Table 1.

Each defect and technique were defined by different optimal regions (variables) carrying discriminant information. It is possible to identify which regions of the specific fingerprints (variables) are the most influential in discriminating between defective and non-defective samples by studying the values of the variable importance in projection (VIP) index. Important variables are indicated by a VIP index greater than one (Wold, Johansson, & Cocchi, 1993).

3.1.2. Mass spectra

In order to avoid differences caused by instrumental analysis over time all mass spectra were row profiled. Logarithmic preprocessing was applied to all of the classes, except fusty and *non-edible*. For these two classes data were autoscaled while the others were only mean-centred before building the final PLS-DA model. Although mass spectrometry is commonly used to provide information about the volatile composition, the use of direct-coupling HS-MS, where the information of the individual compounds is avoided, makes the interpretation difficult. The highest discriminant potential was obtained from 100 to 125 m/z for musty, winey and *defective*, and from 50 to 150 m/z for fusty, rancid and *non-edible* oils. For all of the classes, in particular for musty, winey, fusty and *defective*, a mass of 109 was the one with the highest VIP. Although it is difficult to find the compounds associated with certain masses, the literature indicates that the 109 base

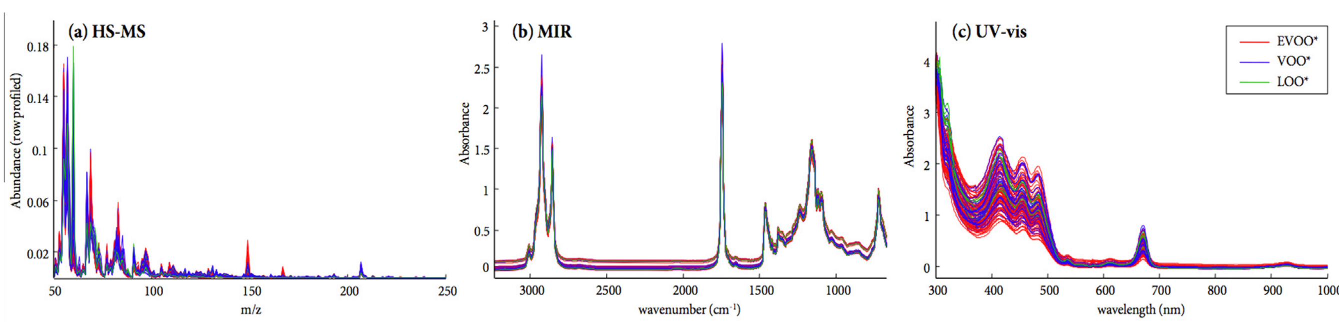

Fig. 2. Instrumental signals obtained by the different techniques to study olive oil samples: (a) HS-MS mass spectra (row profiled); (b) MIR mid-infrared spectra; (c) UV–vis ultra-violet/visible spectra.

Table 1
Optimal PLS-DA models on each instrumental technique for each class studied.

Class	Technique	Region (variables)	Preprocess	Factors (LV)
Non-defective (EVOO)	MS	100–125 m/z	Row profile + autoscaling	2
	MIR	1040–795 cm^{-1}	SNV + 1st derivative + mean-centering	3
	UV–vis	300–1000 nm	Baseline correction (order 3) + mean-centering	4
Non-edible (LOO)	MS	50–150 m/z	Row profile + logarithm + mean-centering	2
	MIR	1040–795 cm^{-1}	SNV + mean-centering	2
	UV–vis	300–1000 nm	Offset correction + mean-centering	2
Musty	MS	100–125 m/z	Row profile + logarithm + mean-centering	2
	MIR	1040–795 cm^{-1}	SNV + 1st derivative + mean-centering	3
	UV–vis	580–1000 nm	SNV + 1st derivative + mean-centering	3
Winey	MS	100–125 m/z	Row profile + logarithm + mean-centering	2
	MIR	1330–1045 cm^{-1}	SNV + 1st derivative + mean-centering	5
	UV–vis	580–1000 nm	SNV + 1st derivative + mean-centering	4
Fusty	MS	50–150 m/z	Row profile + autoscaling	2
	MIR	1040–795 cm^{-1}	Offset correction + 1st derivative + mean-centering	4
	UV–vis	580–1000 nm	Offset correction + 1st derivative + mean-centering	3
Rancid	MS	50–150 m/z	Row profile + logarithm + mean-centering	2
	MIR	3230–670 cm^{-1}	Offset correction + mean-centering	5
	UV–vis	580–1000 nm	SNV + 1st derivative + mean-centering	3

MS: headspace/mass spectrometry, MIR: mid-infrared spectroscopy, UV–vis: UV–vis spectroscopy; SNV: standard normal variate peak is indicative of compounds like guaiacol or 3E-6-methylhepta-3,5-dien-2-one. A few studies have indicated that some key-odorants that contribute to these off-flavoured defective oils are volatile phenols, such as guaiacol, phenol and its methyl, ethyl and vinyl derivatives (Monteleone, 2014; Vichi et al., 2009a, 2009b, 2008). Other studies assign the ketone as a characteristic compound of vinegary and mould defective oils (Purcaro et al., 2014). Other significant musty and winey variables with VIP > 1 were 102, 104, 110, 107, 112 and 119 *m*/*z*. Most of these are characteristic of guaiacol and phenol derivatives, however, some masses can be fragments of other compounds like 1-octen-3-ol, 1-octen-3-one, (E)-2-heptanal and hexanal, related to the musty defect (Monteleone, 2014); and acetic acid and ethyl acetate related to the winey defect (Angerosa, 2002). High VIP scores at 60, 104 and between 54 and 109 *m*/*z* were found in the fusty model. These masses could be related to alcohols, aldehydes, ketones, esters and short chain fatty acids, such as 3-hexenol, 2-hexenal, 2,4-heptadienal, ethyl and butyl acetate, ethyl butyrate, acetic and butanoic acid (Angerosa, 2002; Monteleone, 2014). Finally, rancidity was mostly explained by 50, 72 (VIP > 5) and between 60 and 88 *m*/*z*, which could be attributed to aldehydes, alcohols and acids. In particular aldehydes, such as trans-2-heptenal or the hexanal/nonanal ratio, were suggested as good markers of rancidity and indicators of oxidative status of olive oils (Angerosa, 2002; Kotti, Cerretani, Gargouri, Chiavaro, & Bendini, 2011).

3.1.3. Infrared spectra

Most of the optimal MIR models were pretreated with SNV, except for fusty and rancid where an offset correction was applied. Then, except for rancidity and *non-edible*, first-derivatives were applied, followed by mean-centering of all the resulting spectra before building the PLS-DA model. The olive oil fingerprint region between 1500 and 700 cm^{-1} was included in all of the classification models. Optimal variables for musty, fusty, *non-defective* and *non-edible* belonged to 1040–795 cm^{-1} region, with high VIP values at 1040–1025 cm^{-1}. This zone is assigned to the stretching vibrations of the C–O ester group ($\nu_{C—O}$) which can be attributed to ethyl acetate, butyl acetate, ethyl propanoate or ethyl butyrate that mainly contribute to the fusty perception (García-González & Aparicio, 2010a). Bands around 925–920 and 910–900 cm^{-1} had an important discriminant capacity, however they are difficult to assign. Other common representative wavenumbers were from a region close to 980 and 950 cm^{-1}, due to bending out-of-plane deformation of trans-olefins ($\gamma_{—HC=CH—}$), and a region near 850 cm^{-1}, related to bending vibrations of =CH2 wagging ($\varpi = CH_2$) and out-of-plane deformations ($\gamma_{—C—H}$). These are general rotations, vibrations and deformations that could be assigned to many different organic molecules, and interpreting their correlation with the defects is very difficult. In addition, fusty oils had marked absorbances at 1015 cm^{-1}, often associated with the C–O stretching zone of esters. Another zone from the fingerprint was characteristic of winey-vinegary oils (1330–1045 cm^{-1}). A band at 1115–1100 cm^{-1} related to oleic acid had a higher discriminant power. Other important bands were at 1160, 1135 and 1040 cm^{-1}, belonging to the esters fingerprint zone and associated with stretching vibrations of C–O aliphatic esters, with a strong band at 1160 cm^{-1} due to bending vibrations of CH_2 groups ($\gamma_{—CH_2—}$). The complexity of the matrices makes it difficult to assign the main compounds responsible for the winey defect (acetic acid, ethyl acetate and ethanol) for the experimental spectral bands. Finally, all spectral ranges registered, from 3230 to 2562 and from 2110 to 670 cm^{-1}, were selected to classify rancid oils. Wavenumbers with high VIP values are 2850 and 2920 cm^{-1}, from the CH stretching of cis-double bond vibrations ($\nu_{asymm/symm\ C—O}$) (Guillén & Cabo, 1997), called the fingerprint of oxidation process resulting in a presence of secondary oxidation products, such as aldehydes and ketones. Auto-oxidation is the main cause of rancidity, nonanal and trans-2-heptenal are two examples of the compounds that cause this (Angerosa, 2002; Morales et al., 2005). Other important regions are 1745 and 1155 cm^{-1}, both bands related to ester compounds. The first one is related to C=O stretching vibrations of the ester carbonyl ($\nu_{C=O}$) of triglycerides and free fatty acids and the latter is related to C–O stretching vibrations of aliphatic esters (Guillén & Cabo, 1997).

3.1.4. UV–vis spectra

All of the specific defect models were pretreated with first-derivatives using second-order smoothing polynomials through seven points. Standard normal variate (SNV), offset and baseline (third-order) corrections were applied before the derivatisation, followed by mean-centering of the resulting spectra. From all of the studied variables, the range from 580 to 1000 nm provided the best results to classify the specific defects and the whole spectra (300–1000 nm) was suitable for the general *non-defective* and *non-edible* oils. The first region corresponds to the visible range, which shows the presence of dyes and pigments (Grazia Mignani et al., 2012). A range from 380 to 450 nm, belongs to carotenoid pigments that have high stability. However, chlorophylls and pheophytins, with an exclusive absorption band at 650–700 nm, had a high influence in the model for all of the defects. This may be attributed to the defective samples which have undergone a degradation process, for different reasons, which may have affected the composition of these substances (Sikorska et al., 2007). Another band near 935 nm had significant influence on the classification model for winey and rancid samples. In general, peaks at 610 and 670 nm are reduced for all of the degenerated or defective olive oils.

3.2. Classification results

In order to select the best classification models, all of the data fusion strategies were tested using combinations of some of the optimal regions and pre-processing obtained from the classification achieved by the individual techniques (Section 3.1). All of the data fusion strategies were studied with 135 samples. Twelve samples were initially removed based on individual PCA of fused data where the Hotelling's T^2 and Q residual values were above 95% of the confidence limits.

Each optimal model, for all the data fusion strategies, was selected based on the criteria described in Section 2.4. The optimal results for the conditions selected by all of the data fusion strategies are plotted in Figs. 3 and 4. Figs. 3a and 4a show bar charts of the inaccuracies (%) obtained by test-validation (TV) of the optimum data fusion strategies compared to the individual models. Standard deviations of the 10 replicated models are also plotted. A striped line indicates the lowest inaccuracy obtained by the best individual model for each class studied. The best fusion strategies are shown outside the grey region. Figs. 3b and 4b show the sensitivity vs. specificity Pareto diagram of all the data fusion strategies together with the individual results. Best prediction abilities for each defect or class (balance between selectivity and specificity) are connected by a black line, called *Pareto front*. Final selection of the best classification models for each class and their results, considering inaccuracy, sensitivity and specificity, are summarized in Table 2.

The results obtained by CV (not shown) and TV were similar, ensuring the reliability of the models built. Most of the models are unbalanced (60–70% of samples of one class) and this might cause the specificity, which is the ability to correctly recognize positive samples with defect presence (class with fewer samples), to show higher standard deviations. This is even more noticeable

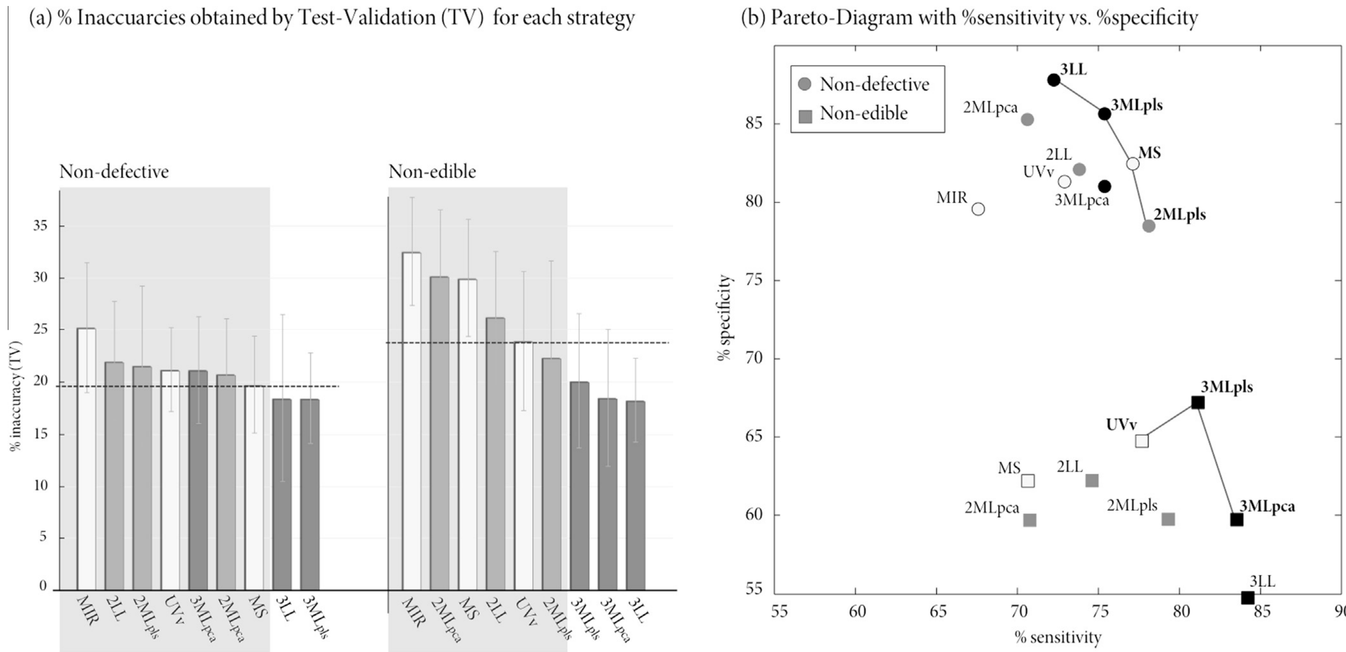

Fig. 3. Final PLS-DA classification parameters (test-validation) for general classes: *non-defective* and *non-edible*. (a) Inaccuracy values (%) for all of the strategies and (b) sensitivity vs. specificity Pareto diagram.

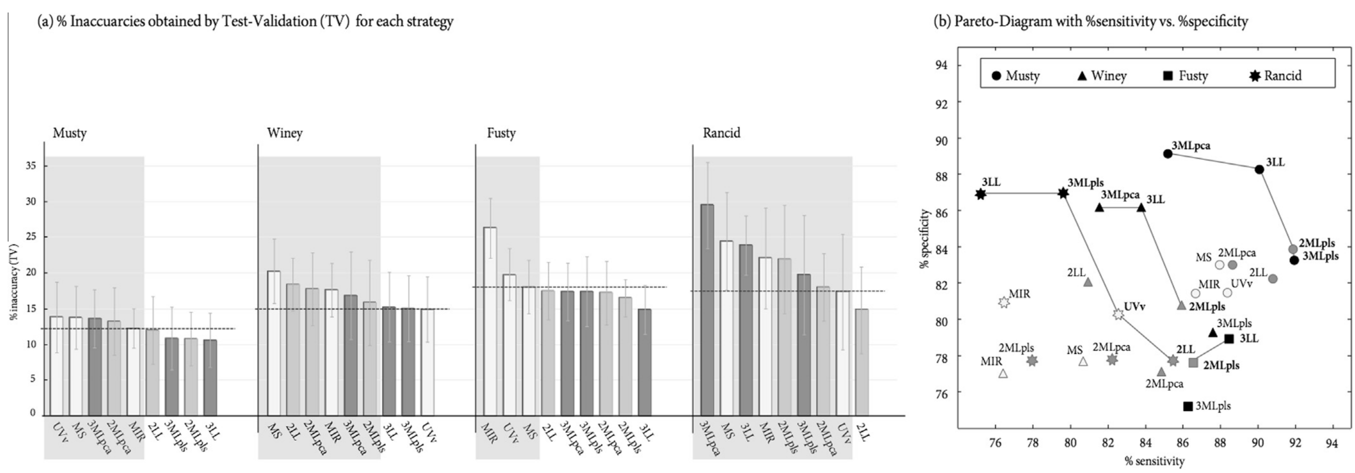

Fig. 4. Final PLS-DA classification parameters (test-validation) for individual defects: *musty*, *winey*, *fusty* and *rancid*. (a) Inaccuracy values (%) for all of the strategies and (b) sensitivity vs. specificity Pareto diagram.

for *non-edible* and *rancid* classifications, due to lack of enough positive samples with these two properties (less than 10%).

To select the best technique or data fusion strategy to classify olive oils based on the presence of defects, the compromise between lower inaccuracy and Pareto results were examined. In most of the models the best classification results were obtained by fusing MS, MIR and UV–vis using low- or mid-level approaches, but in some cases the improvement was not sufficient to justify the use of the three instruments, in such cases MS and MIR or even a single technique being an acceptable option. For all of the defects, mid-level data fusion approaches using PCA scores with two or three techniques did not sufficiently improve (or even worsened) the results obtained by the best individual model.

Non-defective and *non-edible* oil classifications are detailed in Fig. 3, obtaining significantly better results from the *non-defectiveness* classifications.

The best individual technique for *non-defective* oils was MS with inaccuracies of around 20% and sensitivity and specificity above 75%. Compared to data fusion strategies slightly lower inaccuracy results were obtained, using both low- and mid-level fusion of MS, MIR and UV–vis (Fig. 3a – non-defective). In addition, similar sensitivity and specificity to MS were achieved (Fig. 3b). In this case, data fusion was not necessary to improve the results and the best classification was obtained with MS. For the *non-edible* oils the best classifications were achieved by UV–vis spectra (near 25% inaccuracy) and 77/65% (sensitivity/specificity). However, in this case data fusion did improve the best individual results. Inaccuracies were lower in all data fusion strategies using the three techniques (Fig. 3a – non-edible) and the best Pareto results were observed using mid-level data fusion of the three techniques (3MLpls) using PLS-DA scores (Fig. 3b). This approach was the best option, improving sensitivity/specificity from 77/65 to 80/67.5%,

Table 2
Test-validation PLS-DA results of the best individual and data fusion strategies for each class studied (*). Bold and highlighted techniques are the best strategies selected for each class.

Defect	Strategy	Technique	% inaccuracy		% sensitivity		% specificity	
Non-defective (EVOO)	**Individual**	**MS**	19.8	(4.6)	77.0	(7.1)	82.5	(6.6)
	Mid-level PLS-DA (2MLpls)	MS + MIR	21.7	(7.7)	78.0	(5.9)	78.6	(12.3)
	Mid-level PLS-DA (3MLpls)	MS + MIR + UV–vis	18.5	(4.4)	75.3	(7.5)	85.7	(5.6)
Non-edible (LOO)	Individual	UV–vis	24.2	(6.7)	77.4	(4.9)	65.0	(24.1)
	Mid-level PLS-DA (2MLpls)	MS + MIR	22.6	(9.3)	79.1	(13.2)	60.0	(37.6)
	Mid-level PLS-DA (3MLpls)	**MS + MIR + UV–vis**	20.2	(6.5)	81.0	(7.0)	67.5	(26.5)
Musty	Individual	MS	13.7	(4.3)	87.9	(6.8)	83.1	(9.5)
	Mid-level PLS-DA (2MLpls)	MS + MIR	10.7	(3.7)	91.8	(4.5)	83.8	(8.5)
	Low-level (3LL)	**MS + MIR + UV–vis**	10.5	(4.8)	90.0	(6.8)	88.3	(7.0)
Winey	Individual	UV–vis	15.0	(4.5)	90.4	(4.8)	74.3	(10.2)
	Mid-level PLS-DA (2MLpls)	MS + MIR	15.9	(5.9)	85.9	(8.5)	80.7	(9.6)
	Low-level (3LL)	**MS + MIR + UV–vis**	15.4	(4.8)	83.8	(6.2)	86.2	(10.1)
Fusty	Individual	MS	18.0	(3.7)	87.4	(6.6)	71.4	(7.5)
	Mid-level PLS-DA (2MLpls)	MS + MIR	16.6	(2.5)	86.3	(4.6)	77.9	(8.6)
	Low-level (3LL)	**MS + MIR + UV–vis**	14.9	(3.4)	88.1	(5.3)	79.2	(7.3)
Rancid	Individual	UV–vis	17.4	(8.0)	82.9	(8.5)	80.0	(15.8)
	Low-level (2LL)	**MS + MIR**	14.9	(6.0)	85.9	(6.8)	77.5	(18.4)
	Mid-level PLS-DA (3MLpls)	MS + MIR + UV–vis	19.7	(8.3)	79.7	(8.7)	86.7	(17.2)

MS: headspace/mass spectrometry, MIR: mid-infrared spectroscopy, UV–vis: UV–vis spectroscopy
* Results indicated as a percentage and presented as mean (standard deviation) of the 10 models.

respectively, and reducing inaccuracy from 24% (UV–vis) to 20%. To check relevant contribution of all the mid level blocks, the loadings and highest VIP scores were observed. For all the optimized strategies selected, loadings of every data block were considered to be contributing to the final model.

Specific defects, such as mustiness, fustiness, winery and rancidity, are detailed in Fig. 4. The best classification results using individual techniques were obtained with MIR data for musty defective oils with sensitivities and specificities over 80% and around 13% of the samples incorrectly classified. Mustiness showed no significant differences among the three single techniques applied. Winey-vinegary oils had inaccuracies ranging from 15% to 20%, the UV–vis spectra being the one with the lowest value. The highest sensitivity and specificity were 75 and 90%, respectively. Fustiness and rancidity achieved lower inaccuracies (less than 20%) with MS and MIR, respectively.

The classification results of musty, winey and fusty defective oils improved by fusing the three instruments and using low-level data fusion. Mustiness reached lower inaccuracies (Fig. 4a – musty) by fusing two or three techniques with low- and mid-level data fusion, the mid-level strategy always using PLS-DA scores. However, Pareto results (Fig. 4b) indicated that low-level data fusion of all the techniques (3LL) was the best strategy, enhancing sensitivity/specificity from 88/83 to 90/88%, respectively. The inaccuracy was reduced from near 14%, using MS, to 10.5%, using 3LL (Table 2). Winey-vinegary defect achieved similar inaccuracies compared to UV-vis using data fusion (Fig. 4a – winey), but prediction abilities improved when 3LL and 2ML were applied (Fig. 4b). Low-level data fusion of the three instruments was considered the best strategy, improving 75% of specificity obtained by the best individual technique for more than 85%, even though inaccuracy was maintained at around 15% (Table 2). Fustiness reduced its inaccuracies when all of the data fusion strategies tested were used, the 3LL being the best combination (Fig. 4a – fusty). These results were supported by the sensitivity/specificity results (Fig 4b). The inaccuracy was finally reduced from 18% (by MS) to 15% using low-level data fusion of three techniques, and specificity was improved from near 70% to almost 80% (Table 2). In the case of rancidity, an improvement in the classification of the oils was achieved using only MS and MIR low-level data fusion (2LL). Lowering of the inaccuracy of the best individual technique (UV–vis) was achieved (Fig. 4a – rancid), decreasing from 17% to 15% (Table 2); however, sensitivity and specificity showed greatest similarity when PLS-DA (3MLpls) was used rather than the single UV–vis technique or other data fusion strategies, such as mid-level data fusion (Fig 4b). Of all the low-level data fusion approaches, it was found that all the merged blocks had relevance to the final fused model, assuring the complementarity of the data measured by the three instruments. Of all the particular defects, best classifications were obtained for the musty defect. Winey, fusty and rancid defects achieved similar classification abilities using low-level data fusion (at around 15% inaccuracy).

4. Conclusions

The combination of three different instrumental techniques, headspace-mass spectrometry (HS-MS), mid-infrared spectroscopy (MIR) and UV-visible spectrophotometry (UV–vis), can be a useful tool to classify olive oil samples based on their category and the presence of certain sensory defects. The application of different data fusion approaches together with a PLS-DA strategy to select specific regions and pre-processings for each case improved the discrimination capability of the olive oils.

Only in the case of the discrimination of high-quality *non-defective* olive oils (extra-virgin), did data fusion not improve the inaccuracy and sensitivity/specificity values obtained with single MS. For the lowest-quality of olive oils considered *non-edible* (lampante) data fusion of the three techniques (3MLpls) using PLS-DA scores improved individual results. Low-level data fusion was the most suitable strategy to discriminate musty, winey and fusty defects, using HS-MS, MIR and UV–vis, and the rancid defect using HS-MS and MIR. Of all the specific defects, the best classifications were obtained for the musty defect, with inaccuracies of approximately 10% and sensitivity/specificity values near 90%. Winey, fusty and rancid achieved similar classification abilities (around 15% inaccuracy) using low-level data fusion.

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